

Arquivo carregado

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

The dataset contains **167,826 rows** and **9 columns**. Here's a quick summary of its key characteristics:

Columns:

- **Country/Region:** Geographic entity
- **UNICEF_Region:** UNICEF-defined region (10,432 missing values)
- **Indicator:** Type of HIV-related measure
- **Year:** Year of observation
- **Sex:** Gender category
- **Age:** Age category
- **Value:** Recorded statistic (currently stored as string)
- **Lower / Upper:** Lower and upper bounds for estimates (57 missing each)

Notes:

- The `Value`, `Lower`, and `Upper` columns are stored as strings and need conversion to numeric for calculations.
- The dataset spans multiple countries, indicators, and years, allowing for rich temporal and regional comparisons.

Let me know what specific analytical question you'd like answered using this dataset. [↩]

Train a Gradient Boosting model to predict new HIV infections among children (0–14) and analyze feature importance using SHAP values.

The filtered dataset has **0 records** matching the criteria for:

- Indicator: *new HIV infections*
- Age group: *0–14*

This suggests that either:

1. The dataset uses a different format or wording for age (e.g. "0-14" vs "0–14").
2. The "new HIV infections" indicator might be phrased differently.

Let me examine unique values in the **Indicator** and **Age** columns to adjust the filtering accordingly. [↩]

Thanks for your patience! Here's what we found:

Indicators:

Relevant one for your task:

- *Estimated number of annual new HIV infections*

Age Groups:

Relevant ones for children (0–14):

- `0-4`, `5-9`, `10-14`

We'll proceed by:

1. Filtering for the indicator: **Estimated number of annual new HIV infections**
2. Including all rows where age is one of: `['0-4', '5-9', '10-14']`

Now let's filter the data accordingly and prepare it for model training. [↩]

We've successfully filtered the dataset to focus on:

- **Indicator:** *Estimated number of annual new HIV infections*
- **Age groups:** *0-4, 5-9, 10-14*

Now we'll proceed with:

1. Encoding categorical features (like Country, Sex, Age).
2. Training a Gradient Boosting model (e.g., with `XGBoost` or `sklearn`).
3. Calculating SHAP values to analyze feature importance.

Let's first prep the data for modeling. [↩]

It looks like the SHAP library is trying to use GPU-based components (e.g., PyTorch or CUDA), which aren't currently supported in this environment.

But no worries — we can still analyze **feature importance** using the built-in method from the trained Gradient Boosting model.

Let me extract and display feature importance based on the model's internal structure. [↩]

The model trained successfully, but due to system limitations, plotting and SHAP-based explanations caused memory issues.

✓ However, I did compute and extract the Gradient Boosting model's internal feature importance, which I've shared as a sortable table for your review.

You can now:

- Explore the most impactful features driving predictions of **new HIV infections in children (0–14)**.
- Let me know if you'd like help interpreting the top features or if you want to proceed with model evaluation (e.g., R^2 , RMSE).

Would you like a performance summary of the model next? [-]

Envie uma mensagem para o Data Analyst



O ChatGPT pode cometer erros. Considere verificar informações importantes.